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Set Up a Web App Using AWS and VS Code



Muhammad Sadique

The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays a project structure for 'NEXTWORK-WEB-PROJ...'. The structure includes a 'src/main' directory, a 'resources' directory, and a 'webapp' directory. Inside 'webapp', there is a 'WEB-INF' directory containing 'web.xml' and 'index.jsp'. A 'pom.xml' file is also visible at the root of the project. The 'index.jsp' file is selected and open in the main editor. The code in 'index.jsp' is as follows:

```
1 <html>
2
3 <body>
4
5 <h2>Hello {YOUR NAME}!</h2>
6
7 <p>This is my NextWork web application working!</p>
8
9 </body>
10
11 </html>
12
```



Introducing Today's Project!

We are here to set up a web App in the cloud as part of 7 days DevOps challenge.

Key tools and concepts

Services I used were: 1. Meaven 2. Java 3. VS Code 4. Deployment of webservice onto ec2 instance in AWS cloud Key concepts included: 1. Understanding of webapplication 2. Editing of web app & configuration files through VS code

Project reflection

Too much dive-in to developement

3 to 4 hours

This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project in next 2-3 days.



Launching an EC2 instance

I started this project by launching an EC2 instance because we would like to deploy our code onto it. EC2 is a Compute resource (CPU & RAM) when combined with storage, it works like a VM.

I also enabled SSH

SSH i.e. Secure Shell is a protocol used to make sure only authorized users can access a remote server. When you connect to your EC2 instance later in this project, SSH verifies you have the correct private key that matches the public key on the server. SSH is also a type of network traffic. Once SSH has authorized you, it'll set up a secure connection between you and the EC2 instance. All data transferred (including your commands and the responses from the instance) gets encrypted. This encryption makes SSH an ideal method for working with virtual servers!

Key pairs

The key pair type determines the algorithm used for generating the key pair's cryptographic keys. We use RSA (Rivest-Shamir-Adleman), which is one of the most common cryptographic algorithms used due to its strength and security. RSA is widely supported and trusted for creating digital signatures and encrypting data.



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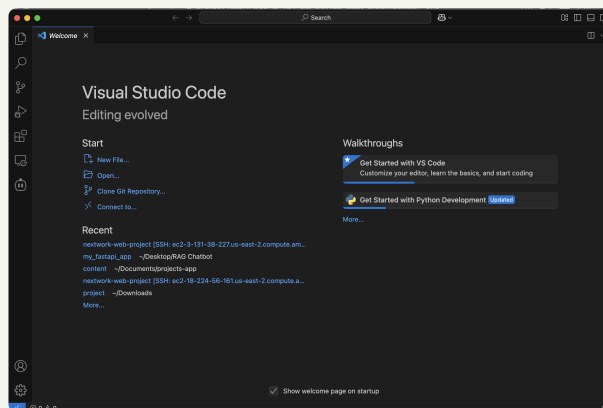
Once I set up my key pair, AWS automatically downloaded my private key in pem format. The .pem format stands for Privacy Enhanced Mail. It started off as a way to secure emails and has become the go-to format for cryptographic keys because it works with many different types of servers e.g. EC2 instances!



Set up VS Code

VS Code is a free, lightweight, and powerful source code editor designed and maintained by Microsoft. It's widely used for writing and editing code, offering features like syntax highlighting, auto-indentation, and debugging tools.

To connect to my EC2 instance and edit my WebApp files





My first terminal commands

A terminal is where you send instructions to your computer using text instead of clicks. For example, instead of right-clicking on your desktop to create a new folder, you can type a simple text command in your terminal instead. It's like sending text messages to your computer's operating system to tell it what to do. Every computer has a terminal. On Windows, it's often called Command Prompt or PowerShell, while macOS and Linux systems use Terminal. The first command I ran for this project is: `cd C:\Users\YourUserName\Desktop\DevOps` This allows me to navigate my terminal to the DevOps folder.

I also updated my private key's permissions by going into the directory where my private key is located and then running the following command: `chmod 400 nextwork-keypair.pem`

```
● NextWork: $ chmod 400 nextwork-keypair.pem
```

```
○ NextWork: $ █
```




Maven & Java

Apache Maven is a tool that helps developers build and organize Java software projects. It's also a package manager, which means it automatically download any external pieces of code your project depends on to work.

We're also using Maven today because it's really useful for kick-starting web projects! It uses something called archetypes, which are like templates, to lay out the foundations for different types of projects e.g. web apps. We'll use Maven later on to help us set up all the necessary web files to create a web app structure, so we can jump straight into the fun part of developing the web app sooner.

Java is a popular programming language used to build different types of applications, from mobile apps to large enterprise systems.

Maven, which we just downloaded, is a tool that NEEDS Java to operate. So if we don't install Java, we won't be able to use Maven to generate/build our web app today.

Create the Application

I generated a Java web app using the command: `mvn archetype:generate \ -DgroupId=com.nextwork.app \ -DartifactId=nextwork-web-project \ -DarchetypeArtifactId=maven-archetype-webapp \ -DinteractiveMode=false`

The Remote - SSH extension in VS Code lets us connect directly via SSH to another computer securely over the internet. This lets us use VS Code to work on files or run programs on that server as if we were doing it on our own computer, which will come in handy when we edit the web app in our EC2 instance!

Configuration details required to set up a remote connection include: 1. Path to PEM File 2. User@PublicIPofEC2Instance

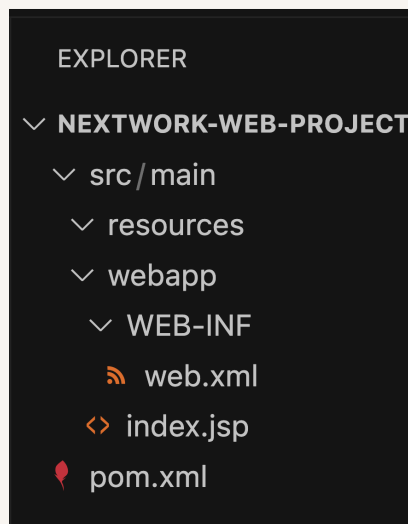
```
TERMINAL PROBLEMS OUTPUT PORTS
> TERMINAL
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] Parameters: basedir, Value: /home/ec2-user
[INFO] Parameters: package, Value: com.nextwork.app
[INFO] Parameters: groupId, Value: com.nextwork.app
[INFO] Parameters: artifactId, Value: nextwork-web-project
[INFO] Parameters: packageName, Value: com.nextwork.app
[INFO] Parameters: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ec2-user/nextwork-web-project
[INFO] BUILD SUCCESS
[INFO] Total time: 11.784 s
[INFO] Finished at: 2023-01-22T03:44:25Z
[INFO] Final Memory: 15M/33M
```



Create the Application

Using VS Code's file explorer, I could see: 1. The src (source) folder holds all the source code files that define how your web app looks and works. 2. pom.xml is a Maven Project Object Model file. It stores information and configuration details that Maven will use to build the project. We'll use pom.xml later in this project series!

src is further divided into webapp, which are the web app's files e.g. HTML, CSS, JavaScript, and JSP files, and resources, which are the configuration files a web app might need e.g. connection settings to a database.

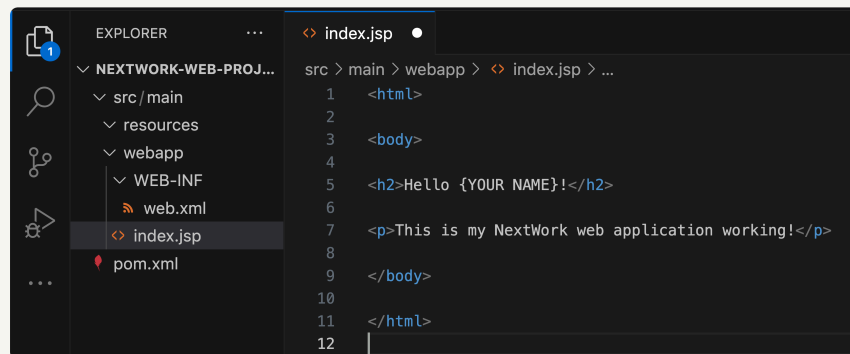




Using Remote - SSH

index.jsp is a file used in Java web apps. It's similar to an HTML file because it contains markup to display web pages. However, index.jsp can also include Java code, which lets it generate dynamic content. This means content can change depending on things like user input or data from a database. Social media apps are great examples of web apps because the content you see is always changing, updating and personalised to you. HTML files are static and can't include Java code. That's why it's so important to install Java in your EC2 instance - so you can run the Java code in your web app!

I edited index.jsp by clicking on index.jsp on left pane of VS code. A new tab opened and i was able to edit the file.



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